



RAJARATA UNIVERSITY OF SRI LANKA
Faculty of Technology

Bachelor of Information Communication Technology
First Year - Semester I Examination – March 2024

ICT 1202 – ELECTRONIC CIRCUITS

Time: Two (02) hours

Answer all Questions.

Calculators will be provided.

There are four (04) questions in three (03) pages.

1. a) Using the Kirchhoff's rules, determine the current through the 10V DC voltage source.

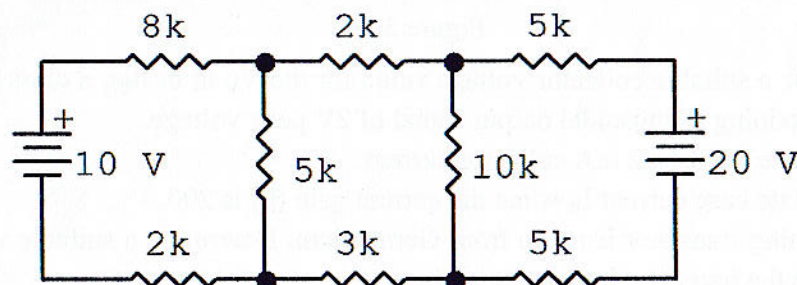


Figure 1

(10 marks)

- b) Explain using the appropriate laws of physics, why an energy band is present in a bulk material instead of a single energy level.

(05 marks)

- c) Using a plot, illustrate the output voltage signal produced by the circuit across AB as shown in figure 2, for a 2V (peak) sinusoidal input voltage signal.

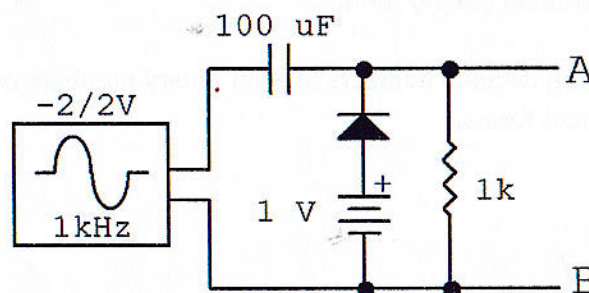


Figure 2

(05 marks)

2. Use the given Transistor circuit to perform the following calculations.

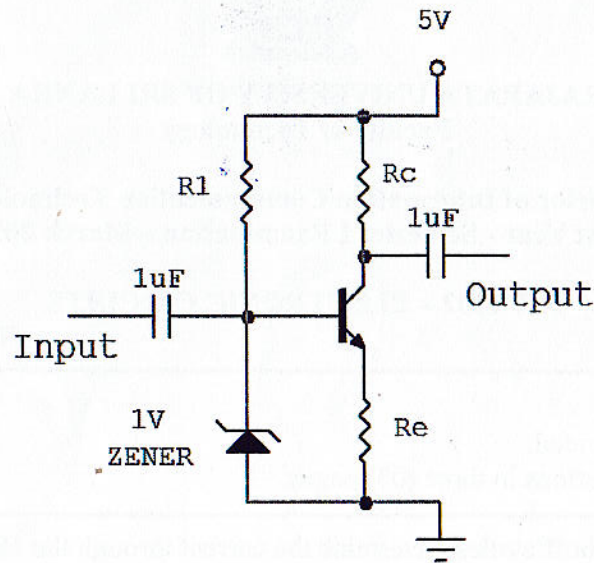


Figure 3

- Choose a suitable collector voltage value for the V_c to design a class A amplifier for obtaining a sinusoidal output signal of 2V peak voltage.
- Calculate R_C for a 2 mA collector current.
- Calculate base current I_B when the current gain (β) is 200.
- Assuming transistor is made from Germanium. Determine a suitable value for R_E to bias the transistor into active region.
- Calculate suitable values for the R_1 .
- What is the purpose of using the Zener diode in this circuit arrangement?

(20 marks)

3) a) Convert the decimal 17.125 number into following forms.

- Binary form
- Octal form
- Hexadecimal form
- Binary coded decimal (BCD) form

(04 marks)

b) Convert the following decimal numbers to 8-bit binary numbers using sign-magnitude and two's complement forms.

- +21
- 19

(04 marks)

Contd.

c) Solve following expressions using Boolean Algebra.

i. $Q = AB + BC(B + C)$

ii. $Q = \overline{ABC} + \overline{A}BC + \overline{AC}$

iii. $Q = (A \oplus B) + ABC + \overline{AB}$

(06 marks)

d) What is meant by the following terms,

i. Digital Logic States

ii. Digital Logic Noise

iii. Digital Logic Gate Noise Immunity

(06 marks)

4) A CCTV camera has following inputs; the camera image (A), an IR motion sensor (B), and the in-built battery voltage level (C). We intend to design an alarm that would go on (play a song) in following conditions. Define 0 as the low level and 1 as the high level.

- Battery voltage level is low.
- Human identification by the camera.
- Motion detection by the IR sensor and human identification by the camera.

a) Construct a truth table for the above system.

(05 marks)

b) Obtain the sum of products, and the product of sums expressions from the truth table.

(05 marks)

c) Construct a Karnaugh Map and obtain a Boolean expression for the results in part (b).

(05 marks)

d) Draw the logic circuit diagram for the Boolean expressions obtained in part (c).

(05 marks)

End.